

TECHNICAL SPECIFICATION
E-Commerce SaaS Platform
=====

This document describes the functional and non-functional requirements of the e-commerce platform as agreed during the discovery phase. Each section corresponds to a distinct deliverable module.

--- Module: ZephyrAuth ---

Description: Authentication and authorisation subsystem.

The ZephyrAuth module is responsible for handling all aspects of authentication and authorisation subsystem. It exposes a dedicated REST API surface and integrates with the core platform via internal service-to-service calls. All ZephyrAuth operations are logged at INFO level for audit trails. Performance SLA: p99 latency < 200 ms at 100 rps.

--- Module: NebulaPay ---

Description: Payment processing engine with stripe integration.

The NebulaPay module is responsible for handling all aspects of payment processing engine with Stripe integration. It exposes a dedicated REST API surface and integrates with the core platform via internal service-to-service calls. All NebulaPay operations are logged at INFO level for audit trails. Performance SLA: p99 latency < 200 ms at 100 rps.

--- Module: AuroraCache ---

Description: Distributed caching layer backed by redis.

The AuroraCache module is responsible for handling all aspects of distributed caching layer backed by Redis. It exposes a dedicated REST API surface and integrates with the core platform via internal service-to-service calls. All AuroraCache operations are logged at INFO level for audit trails. Performance SLA: p99 latency < 200 ms at 100 rps.

--- Module: StellarAPI ---

Description: Public restful api gateway with rate limiting.

The StellarAPI module is responsible for handling all aspects of public RESTful API gateway with rate limiting. It exposes a dedicated REST API surface and integrates with the core platform via internal service-to-service calls. All StellarAPI operations are logged at INFO level for audit trails. Performance SLA: p99 latency < 200 ms at 100 rps.

--- Module: CosmosSearch ---

Description: Full-text search integration via elasticsearch.

The CosmosSearch module is responsible for handling all aspects of full-text search integration via Elasticsearch. It exposes a dedicated REST API surface and integrates with the core platform via internal service-to-service calls. All CosmosSearch operations are logged at INFO level for audit trails. Performance SLA: p99 latency < 200 ms at 100 rps.

--- Module: OrionMetrics ---

Description: Analytics and kpi reporting dashboard.

The OrionMetrics module is responsible for handling all aspects of analytics and KPI reporting dashboard. It exposes a dedicated REST API surface and integrates with the core platform via internal service-to-service calls. All OrionMetrics operations are logged at INFO level for audit trails. Performance SLA: p99 latency < 200 ms at 100 rps.

--- Module: PulsarNotify ---

Description: Real-time notification engine (email + push).

The PulsarNotify module is responsible for handling all aspects of real-time notification engine (email + push). It exposes a dedicated REST API surface and integrates with the core platform via internal service-to-service calls. All PulsarNotify operations are logged at INFO level for audit trails. Performance SLA: p99 latency < 200 ms at 100 rps.

--- Module: VegaExport ---

Description: Data export module - csv, excel, and pdf reports.

The VegaExport module is responsible for handling all aspects of data export module - CSV, Excel, and PDF reports. It exposes a dedicated REST API surface and integrates with the core platform via internal service-to-service calls. All VegaExport operations are logged at INFO level for audit trails. Performance SLA: p99 latency < 200 ms at 100 rps.

The system must handle concu

--- Module: SiriusAdmin ---

Description: Administrative back-office and user management panel.

The SiriusAdmin module is responsible for handling all aspects of administrative back-office and user management panel. It exposes a dedicated REST API surface and integrates with the core platform via internal service-to-service calls. All SiriusAdmin operations are logged at INFO level for audit trails. Performance SLA: p99 latency < 200 ms at 100 rps.

The system must handle co

--- Module: LyraCI ---

Description: Ci/cd pipeline and deployment automation.

The LyraCI module is responsible for handling all aspects of CI/CD pipeline and deployment automation. It exposes a dedicated REST API surface and integrates with the core platform via internal service-to-service calls. All LyraCI operations are logged at INFO level for audit trails.

Performance SLA: p99 latency < 200 ms at 100 rps.

The system must handle concurrent requests from multiple tenants without performance degradation